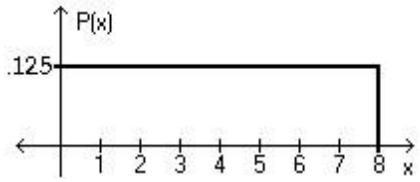
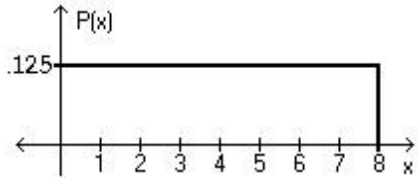
MAT 183 Chapter 6.1 - 6.2 Practice Problems Part-1
Name_____

Using the following uniform density curve, answer the question.

1. What is the probability that the random variable has a value less than 2.5?



2. What is the probability that the random variable has a value between 0.2 and 0.8?

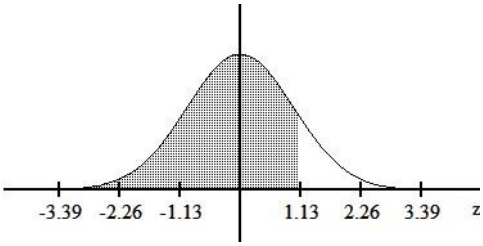


Assume that the weight loss for the first month of a diet program varies between 6 pounds and 12 pounds, and is spread evenly over the range of possibilities, so that there is a uniform distribution. Find the probability of the given range of pounds lost.

3. Less than 11 pounds

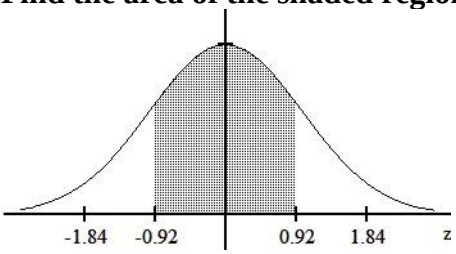
4. Between 8.5 pounds and 10 pounds

5. **Find the area of the shaded region. The graph depicts the standard normal distribution with mean 0 and standard deviation 1.**



Area: _____

6. **Find the area of the shaded region.**

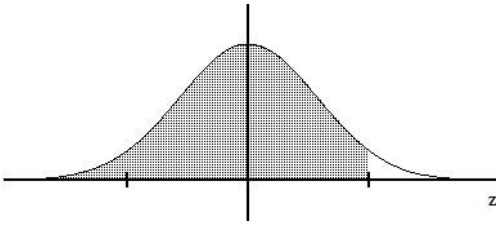


Area:_____

P (-0.92 < Z < 0.92) =----------

Find the indicated z score. The graph depicts the standard normal distribution with mean 0 and standard deviation 1.

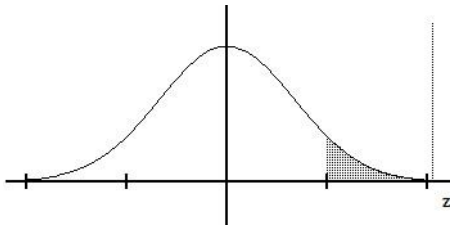
7. Shaded area on the left is 0.9599.



Z = _____

Find the indicated z score. The graph depicts the standard normal distribution with mean 0 and standard deviation 1.

8. Shaded area on the right is 0.0694.



Z = _____

If z is a standard normal variable, find the probability.

9. P(z > 0.59)=

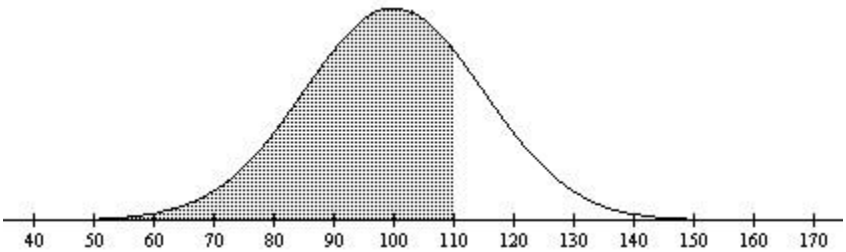
The Precision Scientific Instrument Company manufactures thermometers that are supposed to give readings of **0°C** at the freezing point of water. Tests on a large sample of these thermometers reveal that at the freezing point of water, some give readings below **0°C** (denoted by negative numbers) and some give reading above **0°C** (denoted by positive numbers). Assume that the mean reading is **0°C** and the standard deviation of the readings is **1.00°C.** Also assume that the frequency distribution of errors closely resembles the normal distribution. A thermometer is randomly selected and tested. Find the temperature reading corresponding to the given information.

10. Find P_{96}, the 96th percentile.

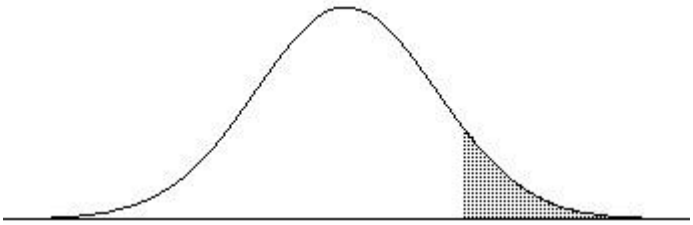
11. Find Q_3, the third quartile.

12. If 7% of the thermometers are rejected because they have readings that are too low, but all other thermometers are acceptable, find the temperature that separates the rejected thermometers from the others.

13. Find the area of the shaded region. The graph depicts IQ scores of adults, and those scores are normally distributed with a mean of 100 and a standard deviation of 15 (as on the Wechsler test).



14. Find the indicated IQ score. The graph depicts IQ scores of adults, and those scores are normally distributed with a mean of 100 and a standard deviation of 15 (as on the Wechsler test).



The shaded area under the curve is 0.10.

Solve the problem. Round to the nearest tenth unless indicated otherwise.

15. In one region, the September energy consumption levels for single-family homes are found to be normally distributed with a mean of 1050 kWh and a standard deviation of 218 kWh. Find P_{45}, which is the consumption level separating the bottom 45% from the top 55%.

16. The amount of rainfall in January in a certain city is normally distributed with a mean of 4.3 inches and a standard deviation of 0.3 inches. Find the value of the quartile Q_1.

Assume that X has a normal distribution and find the indicated probability.

17. The mean is $\mu = 60.0$ and the standard deviation is $\sigma = 4.0$.
Find the probability that X is less than 53.0.

$P(x < 53.0) =$

18. The mean is $\mu = 137.0$ and the standard deviation is $\sigma = 5.3$.
Find the probability that X is between 134.4 and 140.1.

Find the indicated probability.

19. The diameters of bolts produced by a certain machine are normally distributed with a mean of 0.30 inches and a standard deviation of 0.01 inches. What percentage of bolts will have a diameter greater than 0.32 inches?